Xi'an Jiaotong-Liverpool University
西交利物浦大学

Department of Computing

MODULE HANDBOOK

CPT111

Java Programming

Erick Purwanto

Semester 1

2023-2024

SECTION A: Basic Information

❑ Brief Introduction to the Module

Students will learn the basics of programming in Java and in particular Object Oriented Programming. No previous knowledge of programming is assumed. There is a heavy emphasis on practical work and problem solving.

❑ Key Module Information

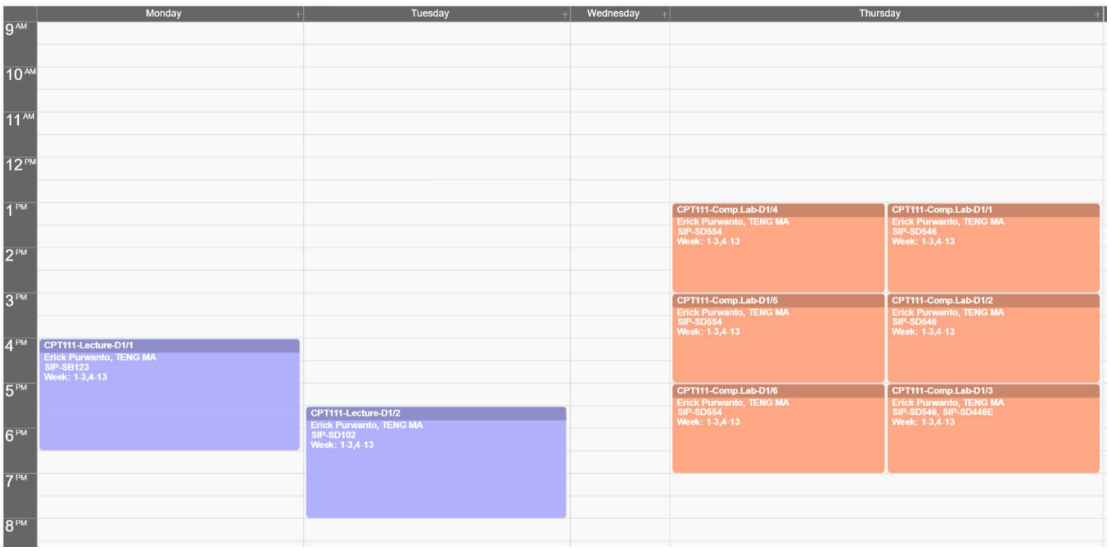
Module name: Java Programming
Module code: CPT111
Credit value: 5
Semester in which the module is taught: 1
Pre-requisites needed for the module: None

Programmes on which the module is shared:
- Information and Computing Science
- Computer Science and Technology
- Digital Media Technology

❑ Delivery Schedule

You will attend one lecture and one lab session per week as per your timetable.

Lecture attendance is compulsory. Lab attendance and participation is graded.



Lectures are timetabled at:

Monday 16:00 – 18:00 SB123 (followed by Q&A, consultation hour)
Tuesday 17:30 – 19:30 SD102 (followed by Q&A, consultation hour)

Lab Sessions are timetabled at:

Thursday 13:00 – 15:00 SD554, SD546
Thursday 15:00 – 17:00 SD554, SD546
Thursday 17:00 – 19:00 SD554, SD546, SD446E

❑ Module Leader and Contact Details

Name: Erick Purwanto

Email address: erick.purwanto@xjtlu.edu.cn

Room number, office hours: SD551, after lectures/by appointment

Preferred means of contact: email

❑ Additional Teaching Staff and Contact Details

Teng Ma teng.ma@xjtlu.edu.cn SD459 Friday, 13:00-15:00

SECTION B: What you can expect from the module

❑ Educational Aims of the Module

The module aims to introduce concepts and principles of problem solving using the computer, and to discuss the construction of appropriate algorithms for problems solving. It also aims to demonstrate principles underlying the design of a high level programming language; and give students experience and confidence in the use of a high level programming language.

❑ Learning Outcomes

On successful completion of this module, students are expected to:

A. Understand and appreciate the principles of using object oriented programming techniques for the construction of professional robust, maintainable programs deployed to meet real world business goals;
B. Design, write, compile, test, debug and execute Java programs using an integrated development environment (IDE);
C. Be able to develop programs that the user can interact with either through graphical representation, or text based menus;
D. Use Java programming to represent, display, and manipulate data as objects while being cognizant of associated risk/safety for safe operation of computing and information systems.
E. Have an awareness of essential legal, social, ethical & professional issues related to software development

❑ Assessment Details

There are 3 coursework components (total **60%** marks) for the module:

Coursework 1 – 15%

Coursework 1 is worth **15%** and is a continuous assessment. This is a weekly auto-graded online coding task, which will make up part of your final grade. You must complete the tasks within 1-2 weeks. You must log into Learning Mall Core to access coursework throughout the semester.

Coursework 2 – 15%

Coursework 2 is a lab attendance and participation every week. This year, we implement Hybrid Escape Room using H5P and Moodle Quiz plus onsite lab activities. Students are grouped into 5-7 group members with 1 Group TA assigned. They all must work together and help each other in completing the lab tasks to get **15%** of the total marks.

Coursework 3 – 30%

Coursework 3 is an individual programming task that you are expected to do in your own time. You can get up to **30%** of the total mark if you fully satisfies the grading rubric. It is planned to be issued in Week 10 (tentative), with a submission date in Week 14 (tentative). Submission will be through either through online autograder or online dropbox on Learning Mall, with details provided later in the semester.

Final Exam - 40%

The exam will take place on exam week. The exam will be onsite (tentative) and open book. We plan for a lab exam, but more details will be provided later in the semester.

❑ Methods of Learning and Teaching

Students will be expected to attend two hours of formal lectures as well as to participate in two hours of supervised labs in a computer lab in a typical week. Lectures will introduce students to the content and practical skills, which are the subject of the module, while computer labs will allow students to practice those skills. The instructors or TAs will lead the completion of lab tasks, but students are expected to try completing the tasks in their own time if they cannot finish them in the lab.

In addition, students will be expected to devote time to solve online ungraded exercises and continuous coding tasks during their private study. Student may ask questions regarding these exercises and coding tasks in the online forum, monitored by the instructors and the TAs. After the deadline, the solutions to these exercises and tasks are accessible online.

Students can seek help and guidance in the online/on-site lab sessions and online forum.

❑ Syllabus & Teaching Plan

Semester	Month	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Description
		28	29	30	31	1	2	3	
Semester 1	September	4	5	6	7	8	9	10	Week 0: Registration and induction
		11	12	13	14	15	16	17	Week 1: Teaching starts
		18	19	20	21	22	23	24	Week 2
	October	25	26	27	28	29	30	1	Week 3; Mid-autumn day; National day
		2	3	4	5	6	7	8	National day; University closure days
		9	10	11	12	13	14	15	Week 4
		16	17	18	19	20	21	22	Week 5
		23	24	25	26	27	28	29	Week 6
	November	30	31	1	2	3	4	5	Week 7
		6	7	8	9	10	11	12	Week 8
		13	14	15	16	17	18	19	Week 9
		20	21	22	23	24	25	26	Week 10
	December	27	28	29	30	1	2	3	Week 11
		4	5	6	7	8	9	10	Week 12
		11	12	13	14	15	16	17	Week 13
		18	19	20	21	22	23	24	Reading Week
		25	26	27	28	29	30	31	Christmas; Reading Week
	January	1	2	3	4	5	6	7	New Year; Examination days
		8	9	10	11	12	13	14	Examination days

Throughout this syllabus, the conceptual aspects of problem solving, algorithm design and the nature of data are expressed alternately with the more specific and implementation-directed aspects of programming and the importance of a reliable, robust and maintainable solution. It is expected that, in teaching the module, the lecturer will further integrate and intermingle these vital topics with the technical aspects of the syllabus.

Topics are listed below:

Week 1 – September 11-15 – Course Admin and Introduction

- Bits & Bytes
- What is Java?
- Data and Variables
- A simple Java program

Week 2 – September 18-22 – Data Types

- Data types
- Scanner/Strings
- Arithmetic and logic operations
- Coding Style
- Cast and convert

Week 3 – September 25-29 – Flow Control

- Boolean and boolean operators
- Branching – if – else – else if
- While Loops

National Holiday Week – October 2-6

Week 4 – October 9-13 – Arrays and For Loops

- Array Definition
- Array Processing
- For Loops

Week 5 – October 16-20 – Strings and Functions

- Characters
- String manipulation
- Static Methods

Week 6 – October 23-27 – Introduction to Objects

- Classes and Objects
- Constructors and Instance Variables
- Object Methods

Week 7 – October 30 - November 3 – Midterm Break

- No new lecture, Midterm Review
- **Hybrid Escape Room 1**

Week 8 – November 6-10 – Object-Oriented Programming

- Inheritance and Overriding
- Polymorphism
- Static Members

Week 9 – November 13-17 – Exceptions

- Exceptions
- Handling Exceptions

Week 10 – November 20-24 – More OOP and Graphics

- OOP Reviews
- Graphics with JavaFX

Week 11 – November 27 - December 1 – File IO and ArrayList

- Input and Output with Files
- Collections
- ArrayList
- Map

Week 12 – December 4-8 – Recursion

- Recursion

Week 13 – December 11-15 – Ethics and Conclusion

- Ethics
- **Hybrid Escape Room 2**

❑ Assessment

Initial Assessment

Sequence	**Assessment Type (EXAM or CW)**	**Method**	**Learning Outcomes Assessed** *(Use codes under Learning Outcomes.)*	**Duration**[3]	**% of Final Mark**	**Resit (Y/N/S)**[4]
#001	CW1, 2, 3	Coursework	ALL		60	S
#002	EXAM	Online Final Exam	ALL	2.0 hours	40	S

Resit Assessment

Assessment Type (EXAM or CW)[5]	**Learning Outcomes Assessed** *(Use codes under Learning Outcomes.)*	**Duration**	**% of Final Mark**
CW	ALL		100

The resit assessment (Resit CW) will assess all of the learning outcomes of the module, and will be weighted as 100% of the final module mark. Other components of the assessment, regardless of whether or not the student passed or failed, will not be included in the calculation of the final module mark, following resit assessment.

- **Reading Materials**

Mandatory (Essential) Textbook: None
(our lecture, lab, and exercise materials are enough)

Optional Textbooks:

Title	Author	Publisher
Introduction to Java Programming and Data Structures, 12th Edition	Y. Daniel Liang	Pearson
Absolute Java, 6th Edition	Walter Savitch	Pearson
A Java Reference	Paul N. Hilfinger	UC Berkeley

SECTION C: Additional Information

- **Student Feedback**

  The University is committed to receiving and responding to student feedback in order to improve the quality of the student experience within the institution. It is University policy that the preferred way of doing this is by using the Online Student Module Feedback Questionnaire Survey. Students are encouraged to complete the questionnaire survey for this module at the end of the semester.

  You are strongly advised to read the policies mentioned below very carefully, because this will help you to perform better in your academic studies. You can find all the policies and regulations related to your academic study from the e-Bridge → 'Quick Reference' → 'Policies and Regulations' page. E-Bridge can be accessed from here and at https://ebridge.xjtlu.edu.cn

- **Attendance**

  The University expects students to attend all timetabled learning sessions associated with this module, and to engage with the relevant learning and support resources. Student attendance will be recorded using the Attendance Management System (AMS). Please follow your teacher's instructions for recording your attendance at each session. Students are responsible for managing their attendance, and should take prompt action to inform the Module Leader in case

circumstances beyond their control affect their class attendance. You are advised to read the University's "Student Attendance Policy" for more information.

- **Rules of Submission for Assessed Coursework**

  The University has detailed rules and procedures governing the submission of assessed coursework. You need to be familiar with the rules and procedures as detailed in the University's "Code of Practice for Assessment".

- **Late Submission of Assessed Coursework**

  The University attaches penalties to the late submission of assessed coursework. You need to be familiar with the rules as detailed in the University's "Code of Practice for Assessment".

- **Mitigating Circumstances**

  The University is able to take into account mitigating circumstances, such as illness or personal circumstances, which may have adversely affected student performance on a module. Students who believe that their performance on an examination or item of assessed coursework may have been impaired by illness or other exceptional circumstances should follow the procedures set out in the University's "Mitigating Circumstances Policy". Such students are also advised to contact their Development Advisor for further guidance and support.

- **Academic Integrity**

  Offences of plagiarism, collusion, copying, submission of commissioned or procured work, and/or the falsification and fabrication of data can result in investigations and penalties being imposed. You need to be familiar with the University's "Academic Integrity Policy".

- **Examination Misconduct**

  The University values academic integrity in both coursework submission and examination conduct. Any examination misconduct will not be tolerated and will result in penalties in accordance with University procedures and regulations as detailed in the "Regulations for the Conduct of Examinations" policy.

- **Generative AI**

  The use of Generative AI for content generation is **not** permitted on all assessed coursework in this module.

  For specific learning activities, students will be explicitly instructed to use Generative AI if and only if it is allowed.

  For more information and resources on Generative AI and your learning and assessment, please consult the "XJTLU AI for Learning" pages of the Learning Mall Core.

- **Learning Mall Core**

  Copies of lecture notes and other materials are available electronically through the Learning Mall Core, the University's virtual learning environment, at: learningmall@xjtlu.edu.cn.

- **Communication**

  All official communication concerning module-related matters will be conducted via e-mail and/or as Learning Mall Core announcements. Other modes of electronic communication are treated as informal.

- **Further Support**

  You are advised to contact your Module Leader in the first instance if you experience any issues with your learning on this module. You may also contact your Academic Advisor or Programme Director. Further information on the kinds of support that the University provides to students can be found in the XJTLU Undergraduate Handbook 2023-24.